

1. Write a Python program for linear Search

```
n=int(input("enter size of a list  "))
a=[int(input()) for i in range(n)]
x=int(input("enter a number to be search  "))
4
for i in a:
    if x==i:
        print(x," value is found in the list ")
```

```
enter size of a list  4
12
56
78
34
enter a number to be search  56
56 value is found in the list
```

2. write a python program for binary search

```
n=int(input("enter size of a list  "))
a=[int(input()) for i in range(n)]
i=0
j=n-1
count=0
x=int(input("enter a value to be search  "))
while i<=j:
    mid=(i+j)//2
    if x==a[mid]:
        count=1
        break
    elif x<a[mid]:
        j=mid-1
    else:
        i=mid+1
if count==1:
    print("value found")
else:
    print("value not found")
```

```
enter size of a list  4
11
22
33
44
enter a value to be search  44
value found
```

3. write a python program for Bubble Sort

```
n=int(input("enter the size of a list"))
a=[int(input()) for i in range(n)]

for i in range(0,n-1+1,1):
    for j in range(0,n-2+1,1):
        if a[j]>a[j+1]:
            temp=a[j]
            a[j]=a[j+1]
            a[j+1]=temp
print("sorted list are ",a)
```

```

enter the size of a list 5
98
87
65
45
33
sorted list are [33, 45, 65, 87, 98]

```

4.write a python program to count occurrence of every single element present in the list

```

pos=[0 for i in range(10)]
neg=[0 for i in range(10)]

n=int(input("enter the size of a list "))
for i in range(0,n-1+1,1):
    x=int(input("enter the values "))
    if x>=0:
        pos[x]=pos[x]+1
    else:
        x=x*-1
        neg[x]=neg[x]+1

print(pos)
print(neg)

print("number","          ","count")

for i in range(10):
    if pos[i]!=0:
        print(i," -----",pos[i])
for i in range(10):
    if neg[i]!=0:
        print(i*-1,"-----",neg[i])

```

```

enter the size of a list 10
enter the values 1
enter the values 2
enter the values 2
enter the values 3
enter the values 3
enter the values 4
enter the values -1
enter the values -3
enter the values -3
enter the values -8
[0, 1, 2, 2, 1, 0, 0, 0, 0, 0]
[0, 1, 0, 2, 0, 0, 0, 0, 1, 0]
number          count
1 ----- 1
2 ----- 2
3 ----- 2
4 ----- 1
-1 ----- 1
-3 ----- 2
-8 ----- 1

```

Double-click (or enter) to edit

